FD CALCULATOR BACKEND

API Request / Response Format

HTTP method	**POST**
Endpoint implemented in this backend	`/api/v1/fd/calculations`
Default port	**8084** (configurable with SERVER_PORT)
Content-Type	`application/json`
Authentication	Bearer JWT required by the current backend configuration
Purpose	Returns a simulation-only FD maturity estimate. It does not open an actual FD account or initiate a payout.

Important: the uploaded backend does not expose `POST /api/fd/calculate`. The implemented route is `POST /api/v1/fd/calculations`. Use the implemented route when sharing API details with your team.

1. Request format

Full local URL (default configuration):

```
POST http://localhost:8084/api/v1/fd/calculations
```

Headers

```
Content-Type: application/json
Authorization: Bearer <JWT_TOKEN>
```

Example JSON request

```
{
  "principal": 100000.00,
  "term": {
    "value": 12,
    "unit": "MONTHS"
  },
  "productCode": "FD001",
  "payoutMode": "CUMULATIVE",
  "startDate": "2026-10-13"
}
```

Request fields

Field	Required?	Format / meaning
principal	Yes	Number in INR; greater than 0; maximum 2 decimal places and up to 10 integer digits.
term.value	Yes	Positive integer representing the duration.
term.unit	Yes	MONTHS or YEARS.
productCode	Yes	Product identifier, e.g. FD001.
payoutMode	No	CUMULATIVE or PERIODIC. If omitted/null, backend defaults to CUMULATIVE.
startDate	No	ISO date YYYY-MM-DD. If omitted/null, backend uses the current date.

Do not send baseRate, customer categories, or bonus values as trusted inputs. The backend documentation states that rates and customer categories are fetched from internal services; client-supplied rate/category values are ignored.

2. Successful response format

A successful request returns HTTP 200 and a JSON object shaped like the example below. Values are illustrative examples, not a guaranteed result for FD001. Actual rates, benefits, APY, maturity date and amounts depend on the product data and customer information returned by the connected services.


```
{
  "calculationId": "CALC-20261013-0001",
  "uuid": "550e8400-e29b-41d4-a716-446655440000",
  "productCode": "FD001",
  "productName": "Fixed Deposit - Short Term",
  "principal": 100000.00,
  "term": {
    "value": 12,
    "unit": "MONTHS"
  },
  "payoutMode": "CUMULATIVE",
  "startDate": "2026-10-13",
  "estimatedMaturityDate": "2027-10-13",
  "interestType": "COMPOUND",
  "compoundingFrequency": "MONTHLY",
  "baseRate": 6.50,
  "categoryBenefits": [],
  "requestedTotalBonus": 0.00,
  "appliedTotalBonus": 0.00,
  "bonusCap": 2.00,
  "effectiveRate": 6.50,
  "apy": 6.70,
  "rateCapped": false,
  "interestEarned": 6700.00,
  "maturityAmount": 106700.00,
  "periodicPayoutAmount": null,
  "currency": "INR",
  "calculationExplanation": "Illustrative explanation text.",
  "generatedAt": "2026-10-13T10:00:00Z"
}
```


Response fields to know

Field	Meaning
calculationId	Human-readable calculation reference.
uuid	Persistent UUID for the saved audit record.
productCode / productName	Product identifier and display name returned for the calculation.
principal / term / startDate	Inputs used for this estimate.
estimatedMaturityDate	Estimated end date calculated from the start date and term.
interestType / compoundingFrequency	Product's interest method and compounding frequency, where applicable.
baseRate / categoryBenefits	Product base rate and customer benefits retrieved by the backend.
effectiveRate / apy	Effective annual rate used and annual percentage yield returned by the calculation.
interestEarned / maturityAmount	Estimated interest and total maturity value in INR.
periodicPayoutAmount	Estimated installment when PERIODIC payout is used; may be null otherwise.
currency / generatedAt	Currency code and timestamp for when the estimate was generated.

3. Expected status codes

HTTP status	Meaning for this endpoint
200 OK	Calculation completed successfully.
400 Bad Request	Malformed JSON or request validation error, such as missing principal or invalid term.
401 Unauthorized	JWT is missing or invalid.
403 Forbidden	Authenticated caller does not have the required CUSTOMER role.
404 Not Found	Product could not be found.
422 Unprocessable Entity	Invalid term, inactive product, or unsupported payout mode.
503 Service Unavailable	Product or Customer service is unavailable.
500 Internal Server Error	Unexpected server-side error.

4. How to test in Postman

1. Select method **POST**.

2. Enter `http://localhost:8084/api/v1/fd/calculations`.

3. Under **Headers**, set `Content-Type: application/json`.

4. Under **Authorization**, choose Bearer Token and supply a valid JWT accepted by your configured environment.

5. Under **Body**, choose raw → JSON and paste the request example from page 1.

6. Send the request and inspect the status code and JSON response.

5. Port and local environment notes

The default server port is 8084, configured in application.yml as `${SERVER_PORT:8084}`. Docker Compose also maps host port 8084 to container port 8084. If the server is running on another machine or in a hosted environment, replace localhost with that machine's reachable hostname or IP and use the configured port.

The current dev profile uses mock Product and Customer clients by default, but the endpoint still has the security requirements declared by the application. A request without a valid JWT may receive 401/403 rather than a calculation response.

OpenAPI / Swagger UI

```
http://localhost:8084/swagger-ui/index.html
http://localhost:8084/v3/api-docs
```

Source of truth: the provided backend ZIP, including FdCalculationController, CalculationRequest, CalculationResponse, application.yml and application-dev.yml. Example response values above are illustrative; the exact runtime response is determined by the configured Product/Customer services and calculation logic.